Tô Gia Bảo

Data Engineer

Thu Duc, Ho Chi Minh City, Vietnam

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Summary

Data Engineer building Python and SQL pipelines, orchestration workflows, cloud data platforms, and conversational analytics tools. At Timo Digital Bank, delivered MCP-powered Slack integrations, operational automation, and analytics solutions with dbt and Lightdash. Hands-on with Airflow, Prefect, Amazon Athena, Amazon S3, PostgreSQL, MongoDB, Docker, and modern AI and lakehouse technologies, with a focus on reliable, maintainable, analytics-ready systems.

Experience

Timo Digital Bank by BVBank — *Ho Chi Minh City, Vietnam*

Data Engineer — *Aug. 2025 – Present*

- Implemented a FastMCP server combining 3 context sources—dbt documentation, Lightdash dashboards, and semantic-layer metadata—to ground conversational analytics in Slack.
- Replaced 2 recurring manual checks—MongoDB backup validation and general ledger lookups—with automated workflows.
- Developed GenBI dashboards with Lightdash, dbt, and Airflow to monitor 3 operational areas: DAG health, Athena query performance, and Amazon S3 usage.
- Created Tracking247, a Streamlit order-tracking application integrated with Airflow, replacing manual refreshes with a daily scheduled pipeline.
- Analyzed operational datasets in Amazon Athena to identify trends, strengthen internal monitoring, and support data-informed decisions.

Projects

VN Market Pulse — *GitHub*

AI Research & Content Platform — *Jul. 2026*

- Built a Vietnamese web-research pipeline that plans LLM queries, searches across Brave, Tavily, Serper, and Exa, deduplicates and reranks sources, extracts bounded context, and generates source-backed Facebook posts.
- Integrated PydanticAI with an OpenAI-compatible Timo gateway for typed output, validation, and usage tracking; added retrieval budgets, retries, and CLI and Streamlit interfaces.

Mini Lakehouse — *GitHub*

Lakehouse & Data Platform — *Jul. 2026*

- Designed a local-first lakehouse with S3-compatible storage, Apache Iceberg, Polaris, Trino, dbt, and Prefect, separating replayable landing data, conformed curated products, and tested analytics marts.
- Engineered contract-driven GitHub Archive and arXiv pipelines with idempotent Iceberg writes; defined catalog, RBAC, and maintenance policies; added dbt tests, documentation, and GLM-OCR on Kaggle GPUs.

Technical Skills

Programming	Python, SQL, C++
Data Engineering	Airflow, Prefect, Dagster, dbt, PySpark, Polars
Lakehouse & Query	Apache Iceberg, Apache Polaris, Trino, Amazon Athena, Amazon S3, PostgreSQL, MongoDB
AI & Integrations	Model Context Protocol (MCP), FastMCP, PydanticAI, Slack, semantic layer design
Analytics & Tooling	Lightdash, Streamlit, Docker

Education

University of Science, VNU-HCM (HCMUS) — *Ho Chi Minh City, Vietnam*

Bachelor of Data Science | GPA: 3.78/4.00 — *2022 – 2026*

Awards & Certifications

2025 **Top 25**, AIC Challenge
2025 **Recipient**, Ninety Eight Scholarship
2024 **Top 10**, RMIT Data Analytics Competition
2024 **Google Business Intelligence**, Professional Certificate
2024 **Google Data Analytics**, Professional Certificate

Publication

GalaxyAssistant: An Intelligent Assistant for Multimedia Event Retrieval — *SOICT*

Co-author — *2025*

Languages

English TOEIC Listening & Reading: 910 | TOEIC Speaking & Writing: 350